

Andrew Qiao

Waterloo, Canada

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Summary

Highly accomplished Computer Science student at the University of Waterloo with extensive experience in developing production-grade AI applications and scalable software systems. Proven track record in leveraging frontier AI technologies, including Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and vector databases to solve complex enterprise problems. Demonstrated expertise in building AI copilots, generative AI applications, and high-performance distributed systems at industry leaders like Snowflake and IBM. Adept at bridging the gap between cutting-edge research and impactful engineering, with a strong foundation in software architecture, CI/CD, and cloud-native development.

Work Experience

- Ramp** Waterloo, Canada
Incoming Software Engineer Intern *Sept. 2025 - Dec. 2025*
 - Selected for upcoming software engineering internship to contribute to high-scale financial technology solutions.
- Snowflake** Waterloo, Canada
Software Engineer Intern *Apr. 2025 - Now*
 - Developing Snowflake's design system consumed by 100+ engineers using React.js, TypeScript, and StyleX.
 - Accelerating the development of polished, accessible, and consistent user experiences across the platform.
- Snowflake** Waterloo, Canada
Software Engineer (Contract) *Not Specified - Not Specified*
 - Built AI Copilot for Notebooks, accelerating workflows for 1,000+ data scientists across 300+ organizations.
 - Owned development of resource monitoring in Notebooks using Python, gRPC, Protocol Buffers, React.js, TypeScript, and Vega Lite.
 - Provided customers with real-time visibility into CPU, disk, and GPU usage through advanced monitoring components.
- Snowflake** Waterloo, Canada
Software Engineer Intern *Not Specified - Not Specified*
 - Designed and implemented an E2E testing framework adopted by 100+ engineers across 15+ teams using Jenkins, Docker, Groovy, Bash, and Datadog.
 - Architected UI component APIs consumed by developers on 5+ teams using React.js and TypeScript.
 - Deployed user experience enhancements across products used by 15,000+ users across 3,000+ organizations.
- IBM** Waterloo, Canada
Software Developer Intern *Not Specified - Not Specified*

- Shipped a generative AI application for code explanation and test-case generation on 12M+ lines of code using AWS, React.js, and IBM's WatsonX AI platform.
- Increased code explanation accuracy from 30% to 90% by implementing Retrieval-Augmented Generation (RAG) using LangChain, FAISS vector database, Python, and the Mixtral 8x7B LLM.
- Engineered a rules-based anomaly detection system to verify \$10B+ in insurance claims using Python and SQL.

- **TD Bank** Waterloo, Canada
Software Developer Intern *Not Specified - Not Specified*
 - Designed a distributed task queue to asynchronously execute portfolio model calculations using Celery and Python.
 - Saved Portfolio Managers 10+ hours per week by eliminating wait times associated with synchronous executions.
- **University of Toronto** Waterloo, Canada
Research Intern *Not Specified - Not Specified*
 - Contributed to a university-partnered study with OpenAI focusing on AI alignment in the workplace.
 - Conducted research under the supervision of Dr. Meena Andiappan and Dr. Gillian Hadfield.

Projects

- **Pethsapp** *Jun. 2021 - Jun. 2023*
Lead Developer
 - Led a team of 5 in developing a mobile application for a high school community, serving 1200+ students, staff, and parents.
 - Utilized React Native, Firebase, Google Cloud, and Sanity for full-stack development.
 - Achieved Top 15 ranking on the Apple App Store Education Chart.

Education

- **University of Waterloo** Waterloo, Canada
Bachelor of Computer Science *2022 - 2026*

Technical Skills

JavaScript, TypeScript, Python, SQL, C++, C, HTML, CSS, React.js, Next.js, Node.js, React Native, Celery, Django, Flask, Tailwind CSS, StyleX, PostgreSQL, SQL Server, MongoDB, Redis, Google Cloud, AWS, Snowflake, Vercel, Datadog, LangChain, FAISS, RAG, LLMs, Docker, Jenkins, gRPC, Protocol Buffers, Vega Lite